Dual-Conditionally Guided Diffusion Models for Fusion of Unregistered Multisource Remote Sensing Images

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***Abstract*— Remote sensing image fusion is a critical technique for enhancing the quality of remote sensing images. Typically, it is presumed that images have been accurately registered to facilitate effective fusion. Traditional methods involve a sequential process of registration followed by fusion, or a parallel approach to registration and fusion, which do not fully leverage the interplay between these two stages. To overcome this limitation, we propose a novel joint learning method termed the dual-conditionally guided registration and fusion diffusion network (RFDifNet) for unregistered remote sensing images. This method integrates image registration and fusion into a unified framework. The RFDifNet comprises two conditional diffusion-based subnetworks: the registration diffusion module (RegDM) and the fusion diffusion module (FusDM). In this architecture, the RegDM corrects the misalignment of the unregistered image and provides it as a conditional input to the FusDM to generate the fused image. Conversely, the fused image is also fed back into the RegDM as a conditional input, enabling a closed-loop iteration of the registration and fusion processes. To further refine the image registration by reducing noise interference and preserving edge details within the RegDM, we introduce a joint learning loss function based on fractional-order derivatives, demonstrating superior performance in geometric and detail preservation compared to traditional methods. Experimental results validate the outstanding performance of the proposed RFDifNet in both image registration and fusion tasks. The source code is available at https://github.com/DDXNJUST/RFDifNet**

***Index Terms*— Conditional diffusion model, image fusion, image registration, multisource remote sensing image.**

I. Introduction

With the development of remote sensing technology, multisource remote sensing image fusion has become an important research direction in remote sensing information processing. Multispectral (MS) and hyperspectral (HS) images contain continuous bands from visible to infrared spectra, which are rich in spectral information and can be used for finer detection and identification than natural images [1], [2], [3]. Panchromatic (PAN) images have clear spatial details and can provide precise surface structure and texture information, which is important for applications such as urban planning, topographic mapping, and disaster monitoring [4], [5]. In addition to spectral–spatial fusion, other necessary fusion paradigms have emerged in remote sensing. The fusion of synthetic aperture radar (SAR) and optical images has attracted attention due to its ability to combine SAR texture features with optical spectral information. However, there are still challenges in maintaining edge consistency [6]. The spatiotemporal fusion technology aims to generate images with high temporal and spatial resolution, and recent advances have been the use of wavelet transform and attention mechanisms to process limited input images effectively [7]. However, in the practice of image fusion in real-world scenarios, there is usually a difficult balance between spatial and spectral dimensions, and it is difficult to directly obtain images that meet high standards in both spatial and spectral dimensions. Relying only on hardware technology upgrades to obtain better quality, high-resolution (HR) HS images is not only expensive but also impractical. Therefore, image fusion methods fuse the spectral features of low-spatial-resolution images and the spatial information of high-spatial-resolution images, which can generate images with both high-spatial-resolution and rich spectral information. Mathematically, the multisource remote sensing image fusion can be formulated as follows:

$$\begin{cases} \mathbf{Y} = \mathbf{CX} \\ \mathbf{Z} = \mathbf{XR} \end{cases} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{MN \times L}$ denotes an image with high spatial and spectral resolution. $\mathbf{C} \in \mathbb{R}^{mn \times MN} (m < M, n < N)$ denotes the spatial degradation, which spatially degrades $\mathbf{X}$ to get $\mathbf{Y} \in \mathbb{R}^{mn \times L}$. $\mathbf{R} \in \mathbb{R}^{L \times l} (l < L)$ denotes the spectral degradation, which subjected $\mathbf{X}$ to a spectral degradation operation to obtain $\mathbf{Z} \in \mathbb{R}^{MN \times l}$. Multisource remote sensing image fusion aims to integrate images from different sensors to overcome their significant differences in radiometric and geometric features. However, reconstructing an approximate $\mathbf{X}$ from $\mathbf{Y}$ and $\mathbf{Z}$ is actually an ill-defined inverse problem [8].

Received 7 February 2025; revised 23 May 2025; accepted 1 June 2025. Date of publication 5 June 2025; date of current version 19 June 2025. This work was supported in part by the Frontier Technologies Research and Development Program of Jiangsu under Grant BF2024070, in part by Jiangsu Geological Bureau Research Project under Grant 2023KY11, and in part by the National Natural Science Foundation of China under Grant 62471235. *(Corresponding author: Liang Xiao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3577073

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Traditional methods usually fuse $\mathbf{Y}$ and $\mathbf{Z}$, but the improvement is limited. HyperNet [9] proposed the HS–MS–PAN three-source fusion framework, which extracts spatial details through multiscale attention modules and preserves spectral information through dense injection blocks. UniNet [10] proposed a unified model, integrated HS–MS, HS–PAN, and HS–MS–PAN, and realized "all-in-one" integration through subnet mutual learning and multiscale gradient attention module, significantly improving practicability.

Moreover, existing image fusion methods impose strict requirements for high-precision registration between source images; otherwise, the disparity can lead to noticeable fusion artifacts. In the traditional paradigm, the registration of multisource images is considered a necessary preprocessing step for image fusion. In this processing framework, the registration and fusion processes are separate, operating independently. When training the proposed network using simulated datasets, high-spatial-resolution image $\mathbf{Z}$ is regarded as a reference registered image. After the transform ϕ, the unregistered image $\mathbf{Z}'$ is obtained. For each pixel point (i, j), the transformation is represented as ϕ:$\mathbf{Z}(i, j) \rightarrow \mathbf{Z}'(i', j')$, resulting in translated coordinates

$$\begin{cases} i' = i + \triangle i \\ j' = j + \triangle j. \end{cases} \tag{2}$$

The objective of registration is to find an inverse transformation ϕ^{-1}:$\mathbf{Z}'(i', j') \rightarrow \hat{\mathbf{Z}}(i, j)$ that maximizes the similarity $\mathcal{C}(\phi^{-1}; \hat{\mathbf{Z}}, \mathbf{Z})$ between registered and reference images. Unsupervised registration aims to maximize the spatial similarity between $\mathbf{Z}'$ and $\mathbf{X}$, identifying the optimal inverse transformation ϕ_*^{-1}.

In recent years, multisource remote sensing image fusion has become a major research hotspot, which can be divided into the following three categories: the fusion of HR PAN and low-resolution (LR) MS images (pan-sharpening) [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], the fusion of HR MS images and LR HS images [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], and the fusion of HR PAN and LR HS images (HS pan-sharpening) [12], [35], [36], [37], [38], [39], [40], [41], [42], [43]. Meanwhile, researchers have developed various registration methods [45], [46], [47], [48], [49], [50], [51], [52], [53]. Recent advances in HS imaging have significantly enhanced target detection capabilities. Traditional methods often treat HS data as 2-D matrices, neglecting intrinsic structural information and limiting detection accuracy. Modern techniques address these limitations through tensor-based processing and spatial–spectral fusion. For instance, TRGLF [54] employs Tucker decomposition with global–local feature fusion to improve target–background separation. In contrast, graph-based methods [55] leverage spectral priors and multiscale spatial features to enhance detection stability under varying illumination conditions. These developments demonstrate the critical importance of jointly modeling spectral features and spatial structure. In essence, image registration is a key preprocessing step in remote sensing image fusion, ensuring that multisource remote sensing images are spatially registered, thus providing an accurate comparison for subsequent fusion and analysis. However, these methods often fail to consider the radiometric and geometric feature differences of images when dealing with multisource remote sensing image registration and fusion, resulting in poor registration and fusion results. The accuracy of the image registration methods directly affects the quality of the fused image, and the fused image must tolerate the errors in the preregistration.

To solve this problem, the proposed method focuses on making image registration and fusion able to avoid mutual negative effects and achieve mutual promotion and joint optimization. We propose a joint learning method termed the dual-conditionally guided registration and fusion diffusion network, called RFDifNet, as shown in Fig. 1.

Compared with traditional methods, the essence of the RFDifNet resides in its ability to mutually promote image registration and fusion, thereby achieving the integration of these two processes. The RFDifNet consists of two diffusion modules, the registration diffusion module (RegDM) and the fusion diffusion module (FusDM), used for image registration and fusion. First, the RegDM performs geometric correction on $\mathbf{Z}'$ unsupervised, enabling $\mathbf{Z}$ to be accurately registered to the reference coordinate system and ensuring spatial consistency. Second, the FusDM combines the registered $\hat{\mathbf{Z}}$ with $\mathbf{Y}$ to generate $\hat{\mathbf{X}}$. Finally, $\hat{\mathbf{X}}$ returns the spatial information to the RegDM. This joint optimization is achieved through fractional-order derivatives, which can more finely describe the local variation characteristics of the function. This method captures rapidly changing details and ensures good geometric preservation. This feedback mechanism is a crucial component of the fusion process, allowing the RegDM to adjust the registered results according to the actual requirements of the fusion. Through the interaction, the fusion process can guide the registration process to focus more precisely on the image regions that significantly impact the final fusion result. In turn, accurate registration performance enhances the fusion results. This dynamic and interdependent relationship ensures the self-optimization of the entire framework. Overall, the process forms a closed-loop system. The system continuously iterates and optimizes to achieve better fusion and registration results.

The main contributions are as follows.

1) We propose the RFDifNet, the first framework to unify image registration and fusion using dual-condition-guided diffusion models (RegDM and FusDM). Unlike existing methods that handle these tasks separately, our method achieves geometric consistency fusion by utilizing the RegDM to provide precise registration guidance. At the same time, the FusDM adaptively reduces modal differences. This joint optimization significantly enhances these two tasks collaboratively.
2) The RFDifNet introduces a bidirectional interaction mechanism between the RegDM and the FusDM. The registration process continuously improves spatial alignment to enhance fusion quality, while the fusion output provides feedback to optimize registration further. This dynamic interaction achieves mutually enhanced generation through conditional diffusion, resulting in better performance.

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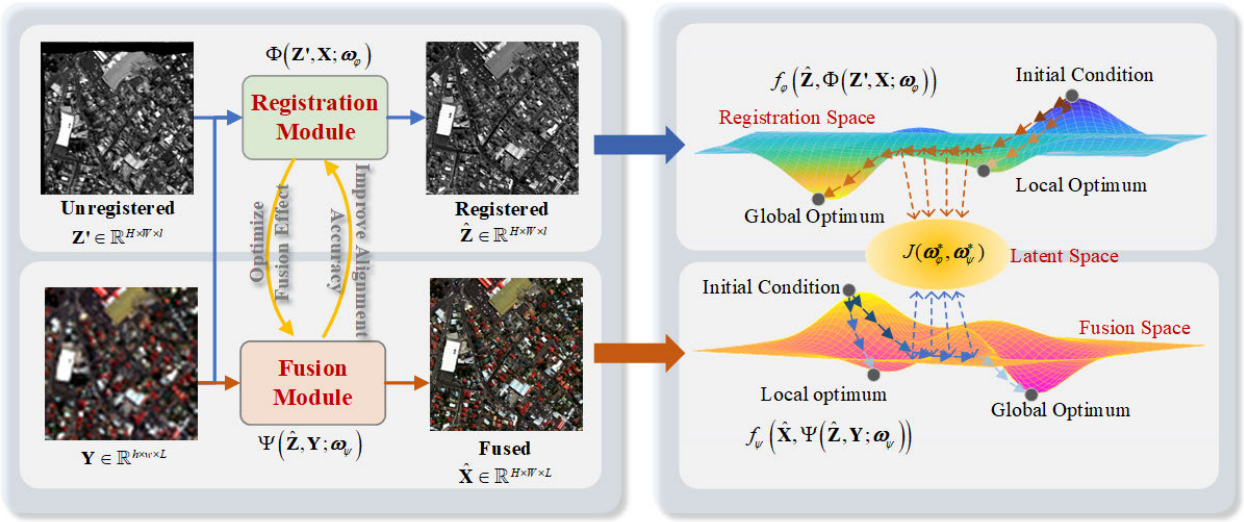


Fig. 1. Correspondence between the application framework and problem paradigms of the RFDifNet. (As shown in the figure, only PAN and MS images are taken as an example for demonstration, but the proposed RFDifNet can be applied to all image fusion tasks.) The unregistered image pairs, denoted as $[\mathbf{Z}', \mathbf{Y}]$, are concurrently fed into the registration module $\Phi(\mathbf{Z}', \mathbf{Y}; \omega_\varphi)$ and the fusion module $\Psi(\hat{\mathbf{Z}}, \mathbf{Y}; \omega_\psi)$. This parallel input process gives rise to the generation of the registered image $\hat{\mathbf{Z}}$ and the fused image $\hat{\mathbf{X}}$. An interactive optimization mechanism is established between Φ and Ψ. Module Φ plays a facilitating role in improving the quality of the fusion outcome, while module Ψ functions to augment the precision of the registration module. ω_φ and ω_ψ represent the learnable parameters. The objective of the proposed method is to fully leverage this framework for the pursuit of optimal parameters $[\omega_\varphi^*, \omega_\psi^*]$. By it, a global optimum can be attained in the latent space [57]. This is achieved through the utilization of the constrained term $J(\cdot)$ that enables the joint optimization of the two tasks. Herein, $f_\varphi(\cdot)$ and $f_\psi(\cdot)$ denote the fidelity terms. (a) Application framework. (b) Problem joint optimization.

3) We introduce a new joint loss based on fractional-order derivatives to guide the image registration further. Due to their smoothing properties, fractional-order derivatives are gentler than integer-order derivatives in dealing with high-frequency components. They can depict the details of the image more finely, especially in reducing noise effects and preserving the texture information of the image, thus improving the quality of image registration.

II. Related Work

A. Image Fusion

1) Fusion of MS and PAN Images (Pan-Sharpening): Traditional pan-sharpening methods include component constitution (CS)-based methods such as intensity–hue–saturation (IHS) transform [11], principal component analysis (PCA) [12], and Gram–Schmidt (GS) transform [13]. Although this method has high computational efficiency, it is prone to introducing spectral distortion. Another kind of method is based on multiresolution analysis (MRA), which uses methods such as wavelet transform [14], contourlet [15], and curvelet [16] to extract spatial details from the PAN image and inject them into the MS image. The MRA method may not be as significant as the CS method in enhancing spatial details. In addition, degradation model-based methods consider pan-sharpening as an image recovery problem [17], assuming that the LR MS and PAN images are different degraded forms of the HR MS image, respectively. This method relies heavily on the accuracy of the degradation model, thus its performance is inherently limited.

In recent years, significant advancements have been made in deep neural networks (DNNs). Generative adversarial networks (GANs) have been employed to improve image fusion quality. Diao et al. [19] proposed the zero-reference ZeRGAN. UCGAN [20] uses an unsupervised learning paradigm. The training process of these methods may be unstable and difficult to control. Transformer architecture has also been introduced into this field. The HyperTransformer proposed by Bandara and Patel [21] has achieved good results by fusing texture and spectral features, but its computational complexity is relatively high. The latest development trend is based on diffusion model methods, such as Meng et al.'s [22] PanDiff and Cao et al.'s [23] DDIF. These methods can generate high-quality results through a gradual denoising process and perform well in feature preservation, but their inference speed is relatively slow. Balancing computational efficiency and result quality is still a problem that needs to be solved.

2) Fusion of MS and HS Images (HS–MS): The fusion of MS and HS images encompasses extending pan-sharpening-based methods, matrix factorization, tensor representations, and deep learning methods. Pan-sharpening can be applied by selecting the HR MS spectral band to enhance the spatial resolution of LR HS images [24]. This method has high computational efficiency but is prone to spectral distortion. Matrix factorization methods obtain high-fidelity images by decomposing HS images into low-dimensional matrices and using a reconstruction model optimized with low-rank priors [25], [26]. Although they can maintain spectral characteristics, they are more noise-sensitive. Tensor representation methods employ low-rank decomposition methods and leverage low-rank priors [29]. Outstanding performance in maintaining structural information, but high computational complexity.

Deep learning methods learn the complex features of images by training DNNs to learn the mapping from LR to HR. PSRT [30] uses a pyramid architecture and a Transformer

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to improve the quality of fused images. The MSST-Net [31], based on the Swin Transformer, extracts multiscale features. For the fusion of unregistered HS images, MoE-PNP [32] proposes the principle of registration-fusion consistency, enhancing the interpretability of the fusion process. Qu et al. [33] proposed a registration-fusion co-optimization network and its progressive and iterative design improves the fusion accuracy. In addition, Liu et al. [34] adapted the diffusion model to spectral data and then transferred its spectral distribution knowledge to introduce a regularization term. However, existing deep learning methods generally suffer from strong dependence on training data and insufficient physical interpretability.

3) Fusion of HS and PAN Images (HS Pan-Sharpening, HS–PAN): In recent years, with the widespread application of HS systems, traditional pan-sharpening methods have been applied to improve the resolution of HS images. Methods based on the CS, such as PCA [12] and its improved version GFPCA [36], achieve fusion by replacing or enhancing specific bands. However, the modeling ability for interband correlation is limited. Methods based on MRA enhance the spatial resolution of PAN images and HS images by decomposing them at different scales and then injecting the high-frequency detail information of the PAN images into the HS images [37], [38]. However, spectral distortion may be introduced in high-frequency information fusion. Bayesian-based methods treat the fusion process as a Bayesian inference problem and optimize the results by setting reasonable prior probabilities [39]. Strong flexibility, but high computational complexity, and dependence on prior rationality.

Deep learning-based methods have become a research hotspot in recent years due to their powerful nonlinear mapping ability. Bandara et al. [40] proposed a method combining an improved deep image prior and residual reconstruction. DHP-DARN [41] uses a deep prior and a dual-attention residual network to enhance the information fusion ability. HyperPNN [42] uses a spectrally predictive convolutional neural network to learn the relationships among bands. The FPFNet [43] proposes a feature pyramid fusion network to extract and fuse features at multiple scales. Rui et al. [44] developed a model that combines the pretrained diffusion model and the generalization ability of Bayesian methods to capture the spatial structure of images. It achieves excellent performance in spatial structure modeling, but suffers from low inference efficiency. Overall, existing methods must be further balanced regarding complexity and other aspects.

B. Image Registration

Image registration is a challenging task in remote sensing, mainly due to geometric differences, intensity differences, and noise. Intensity-based methods estimate the transformation relationship by utilizing the pixel intensities of the reference image and the image to be registered [45]. This method is computationally efficient but sensitive to geometric deformation and radiation differences. Feature-based methods estimate transformation relationships by extracting and matching important image features [46], [47]. They can effectively handle geometric and radiometric differences, but due to the robustness of feature extraction, their performance is limited in scenes with texture loss or severe noise. SIFT [56] is a benchmark method for traditional registration, achieving image registration by detecting and describing scale and rotation-invariant local features. However, when faced with the multimodal characteristics of remote sensing images, the accuracy of descriptor matching may decrease. Intensity-feature-based methods combine the advantages of intensity-based and feature-based methods, but there is still a tradeoff between computational complexity and real-time performance [48], [49].

The deep learning-based method automatically learns advanced semantic features through neural networks, significantly improving the generalization ability of registration. The CHA-Net [50] solves multimodal registration problems through domain adaptation, but it relies on a large amount of unlabeled data and has high training costs. TransMorph [51] is a Transformer-based architecture that captures long-range spatial dependencies through self-attention mechanisms, making it particularly effective for large deformations. DiffuseMorph [52] utilizes a diffusion model to describe registration as a progressive deformation process, demonstrating strong performance in unsupervised deformable registration. FSDiffReg [53] guides with the diffusion model from the feature and score dimensions to achieve precise registration. The method based on the diffusion model performs well in complex deformations, but the inference speed is slow.

C. Diffusion Model

The diffusion model is one of the generative models in deep learning. It generates new data samples by simulating the diffusion process of data distribution. The fundamental derivation of the diffusion model is outlined below.

1) Forward Diffusion: First, a continuous time series $t \in [0, T]$ is defined. The process represents the transition from the initial data x_0 to the final noise x_T. The diffusion model assumes that this process can be represented as a series of Markov processes. Among them, x_t represents the data state at time t. The forward diffusion process can be represented as

$$q(x_{1:T}\,|x_0) = \prod_{t=1}^{T} q(x_t\,|x_{t-1}) \tag{3}$$

where $q(x_t|x_{t-1})$ is the transfer probability, which is usually assumed to be Gaussian distributed

$$q(x_t\,|x_{t-1}) = \mathcal{N}\left(x_t\sqrt{1-\beta_t}x_{t-1}, \beta_t I\right) \tag{4}$$

where β_t is the variance schedule, which determines the amount of noise added at each step, and I is the unit matrix.

2) Reverse Diffusion: The reverse process involves reconstructing the original data x_0 from the noise x_T by learning a model $p_\theta(x_{0:T}|x_T)$ that represents this reverse process. To parameterize the model $p_\theta(x_{0:T}|x_T)$, we typically assume that it follows a Gaussian distribution:

$$p_\theta\,(x_{t-1}\,|x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta\,(x_t, t)\,, \sigma_\theta^2\,(x_t, t)\right) \tag{5}$$

where the mean μ_θ and variance σ_θ^2 can be predicted by the neural network. θ denotes the model parameters.

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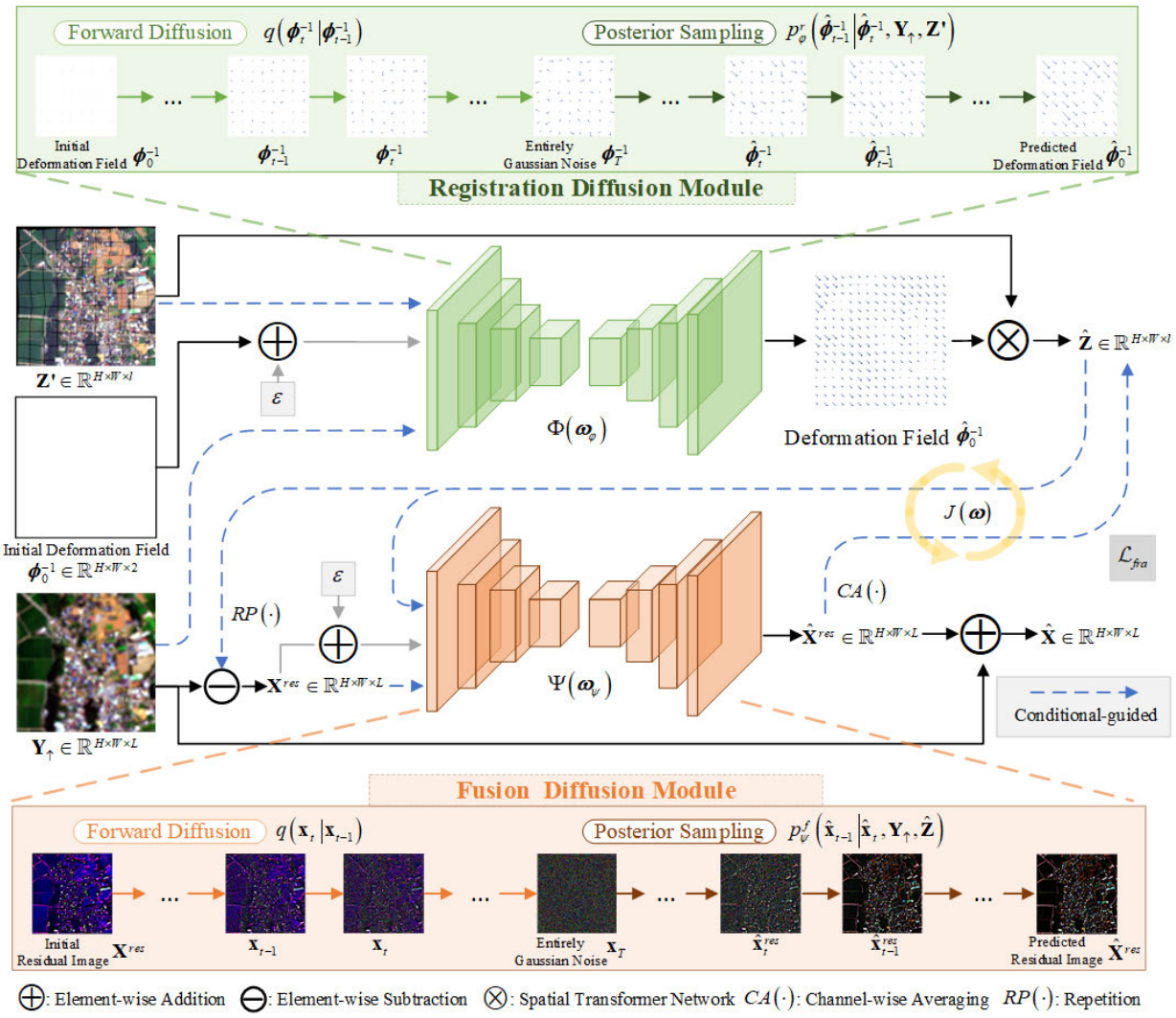


Fig. 2. Schematic of the overall framework of the RFDifNet. It consists of the RegDM and the FusDM. CA(·) represents the channel-wise averaging operator, which computes the mean across channels and broadcasts the result back to the specific tensor shape.

3) Sampling: When generating new samples, we begin with x_T and progressively apply the learned inverse process to reconstruct the samples, denoted as $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$, until $t = 0$. This method enables the generation of new samples that accurately reflect the distribution of the training dataset.

4) Conditional Diffusion Model: The conditional diffusion model builds on the standard diffusion model by integrating conditional variables to steer the generation of samples. In the inverse process, the objective is to gradually recover the original data x_0 from the pure noise x_T. For the conditional diffusion model, this process depends not only on the current state x_t but also on the conditional information y. As a result, the conditional probability distribution

$$p_\theta(x_{t-1}|x_t, y) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t, y), \sigma_\theta^2(x_t, t, y)\right) \quad (6)$$

where μ_θ and σ_θ^2 are parameterized by the neural network and depend on the current state x_t, the time step t, and the conditional information y. The conditional diffusion model, by incorporating conditional information y, enables the generative process to produce more diverse and precise results based on specific conditions.

III. Image Alignment and Fusion Co-Optimization Method

A. Problem Formulation

The two tasks are interdependent, and we integrate the forward diffusion processes associated with them into a unified joint process. Consider two sequences of random variables $\{x_t\}$ and $\{y_t\}$. The joint probability distribution of the forward diffusion can be expressed as follows:

$$q(x_{1:T}, y_{1:T}|x_0, y_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}, y_{t-1}) q(y_t|x_{t-1}, y_{t-1}). \quad (7)$$

We aim to learn a model $p_\theta(x_{0:T}, y_{0:T}|x_T, y_T)$ to represent the joint reverse process. We will parameterize the conditional probability as a Gaussian distribution for each period. For each

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time step t, we will parameterize the following conditional probabilities:

$$p_\theta\left(x_{t-1}\,|x_t, y_t\right) = \mathcal{N}\left(x_{t-1}; \mu_{x,\theta}\left(x_t, y_t, t\right), \sigma^2_{x,\theta}\left(x_t, y_t, t\right)\right) \tag{8}$$

$$p_\theta\left(y_{t-1}\,|x_t, y_t\right) = \mathcal{N}\left(y_{t-1}; \mu_{y,\theta}\left(x_t, y_t, t\right), \sigma^2_{y,\theta}\left(x_t, y_t, t\right)\right). \tag{9}$$

Among them, $\mu_{x,\theta}(x_t, y_t, t)$ and $\mu_{y,\theta}(x_t, y_t, t)$ are mean functions, and $\sigma^2_{x,\theta}(x_t, y_t, t)$ and $\sigma^2_{y,\theta}(x_t, y_t, t)$ are variance functions. All of these functions are determined by the parameter θ. They are designed to capture the complex relationships and patterns that exist between the variables as we trace back in time during the reverse process. During the training phase, we employ an alternating optimization strategy. Specifically, we iteratively update generative model parameters related to x and y. This method allows the model to learn the unique characteristics of each task and their interdependent relationship in a balanced manner.

The RFDifNet can jointly optimize image registration and fusion. The objective of the RFDifNet is to identify the optimal parameters for both tasks. The RFDifNet assumes that the input unregistered images are of the sizes $H \times W$, denoted as unregistered image pairs $\mathbf{Z}' \in \mathbb{R}^{H\times W\times l}$ and $\mathbf{Y} \in \mathbb{R}^{h\times w\times L}$, respectively, to obtain the registered image $\hat{\mathbf{Z}} \in \mathbb{R}^{H\times W\times l}$ and fusion image $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times L}$. As shown in Fig. 1, the RFDifNet represents the tasks of registration and fusion into a single objective, which can be formulated as an optimization model

$$\min_{\omega_\varphi,\omega_\psi} f_\varphi\left(\hat{\mathbf{Z}}, \Phi\left(\mathbf{Z}', \mathbf{X}; \omega_\varphi\right)\right) + f_\psi\left(\hat{\mathbf{X}}, \Psi\left(\hat{\mathbf{Z}}, \mathbf{Y}; \omega_\psi\right)\right) + J\left(\omega_\varphi, \omega_\psi\right). \tag{10}$$

Among them, the registration function $f_\varphi(\cdot)$ and the fusion function $f_\psi(\cdot)$ have learnable parameters ω_φ and ω_ψ. $J(\cdot)$ is a fidelity term that jointly optimizes the two tasks by designing a loss that preserves the structural features of the image.

As shown in Fig. 2, the proposed RFDifNet combines the problem paradigm with joint optimization learning. First, the RegDM is employed to construct the registration condition guidance based on the spatial features of $\mathbf{Y}_\uparrow$, resulting in a preliminarily registered image $\hat{\mathbf{Z}}$. Next, the FusDM performs image fusion by utilizing the roughly registered $\hat{\mathbf{Z}}$ and the low-spatial-resolution image $\mathbf{Y}_\uparrow$ as guidance conditions. This module outputs the fused image by extracting the residual features of both spatial and spectral information. Subsequently, a joint optimization loss based on the fractional-order derivative is utilized to extract the residual features and conduct fine corrections on $\hat{\mathbf{Z}}$. As a nonlocal operator, the fractional-order derivative effectively captures texture details and edge information, enhancing registration accuracy. Joint optimization loss is a conditional guidance term used to optimize the RegDM. Throughout the training process, the RegDM and FusDM implement incremental corrections and alternating optimizations. The training process is continuously monitored through the validation dataset. When the number of iterations reaches 300 000, the network performance tends to stabilize, and continuing training at this time cannot bring significant performance improvement. So, choose to terminate when the number of iterations is 300 000 to ensure obtaining high-quality registration and fusion images. The network architectures of the RegDM and FusDM are based on U-Nets with an attention mechanism.

B. RegDM

Complex geometric distortions are common in remote sensing image analysis, making image registration and subsequent processing complex. To address this issue, the RFDifNet introduces an unsupervised RegDM aimed at achieving spatial registration between two remote sensing images using a diffusion model without a reference image. However, traditional diffusion models have an inherent issue in the image generation process: a lack of control over the generation process, which makes it difficult to guide the model to generate images with specific content. To overcome this limitation, the RFDifNet employs a conditional generation method, utilizing $\mathbf{Y}_\uparrow$ to guide the registration process of $\mathbf{Z}'$. The conditional diffusion model can provide outputs with enhanced spatial details while maintaining the generated deformation field.

During the forward diffusion process, at time step t, Gaussian noise is gradually added to the deformation field $\boldsymbol{\phi}_\mathbf{0}^{-\mathbf{1}}$ with an initial zero matrix. For each step $t \in [0, T]$, the sampling of latent variables is defined as a Gaussian transition $q(\boldsymbol{\phi}_\mathbf{t}^{-\mathbf{1}}|\boldsymbol{\phi}_{\mathbf{t-1}}^{-\mathbf{1}})$ until $\boldsymbol{\phi}_\mathbf{T}^{-\mathbf{1}}$ becomes entirely Gaussian noise. In the posterior sampling process, at step t, the input of Φ consists of $\hat{\phi}_t^{-1}$, the unregistered image $\mathbf{Z}'$, and the spatial degradation $\mathbf{Y}_\uparrow$ of image $\mathbf{X}$, formulated as $\mathbf{z}_{t-1}^{\text{in}} = \{\hat{\phi}_t^{-1}, \mathbf{Y}, \mathbf{Z}'\}$. Subsequently, Φ outputs a deformation field $\hat{\phi}_{t-1}^{-1}$. Ultimately, $\hat{\boldsymbol{\phi}}_\mathbf{0}^{-\mathbf{1}}$ is obtained. Then, by inputting $\mathbf{Z}'$ and $\hat{\phi}_0^{-1}$ into the spatial transformer network (STN) [58], the predicted registered image $\hat{\mathbf{Z}} = \Phi(\mathbf{z}^{\text{in}}, t; \omega_\varphi)$ is generated.

To train RegDM in an unsupervised paradigm, diffusion loss is employed to constrain the generation process

$$\mathcal{L}_{\text{Diff}}^R\left(\omega_\varphi\right) = \mathbb{E}\left\|\Phi\left(\mathbf{z}^{\text{in}}, t; \omega_\varphi\right) - \varepsilon_r\right\|_2^2, \quad \text{s.t.} \quad \varepsilon \sim \mathcal{N}(0, I) \tag{11}$$

where ε_r denotes Gaussian noise. In addition, a regular smoothness penalty is applied to the deformation field using the parameterized deformation field $\hat{\phi}_0^{-1}(\omega_\varphi)$ loss

$$\mathcal{L}_\phi^R(\omega_\varphi) = \sum\left\|\nabla\hat{\boldsymbol{\phi}}_\mathbf{0}^{-\mathbf{1}}(\omega_\varphi)\right\|_F^2. \tag{12}$$

Finally, normalized mutual correlation loss is employed to maximize the spatial similarity between $\hat{\mathbf{Z}}$ and $\mathbf{Y}_\uparrow$. This loss function evaluates the similarity within local regions between the input image and the target image by calculating the local normalized cross correlation (NCC)

$$\mathcal{L}_{\text{NCC}}^R(\omega_\varphi) = \frac{\sum_{i,j}\left(\hat{\mathbf{Z}}(i,j) - \bar{\hat{\mathbf{Z}}}\right)\left(\mathbf{Y}_\uparrow(i,j) - \bar{\mathbf{Y}}_\uparrow\right)}{\sqrt{\sum_{i,j}\left(\hat{\mathbf{Z}}(i,j) - \bar{\hat{\mathbf{Z}}}\right)^2\sum_{i,j}\left(\mathbf{Y}_\uparrow(i,j) - \bar{\mathbf{Y}}_\uparrow\right)^2 + \epsilon}} \tag{13}$$

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where $\hat{\mathbf{Z}} = \Phi(\mathbf{z}^{\text{in}}, t; \omega_\varphi)$, $\bar{}$ represents the mean, and ϵ represents the minimum value. The total loss function of the RegDM is defined as

$$\mathcal{L}^R(\omega_\varphi) = \mathcal{L}^R_{\text{Diff}}(\omega_\varphi) + \alpha\mathcal{L}\phi^R(\omega_\varphi) + \beta\mathcal{L}^R_{\text{NCC}}(\omega_\varphi) \tag{14}$$

where α is empirically set to 20. β is set to 200.

C. FusDM

After the RegDM, the RFDifNet constructs the FusDM. The proposed method uses a fusion method to generate more accurate fusion results from multisource images. Even if there are specific errors in the image registration process. However, traditional fusion methods exhibit limitations in practical operations. Their stability and adaptability are relatively poor, which makes it difficult to accurately perform fusion operations when processing various images from different sources. The FusDM introduces a conditional diffusion model. By using $\mathbf{Y}_\uparrow$ and the registered $\hat{\mathbf{Z}}$ as the conditional input, it provides precise and targeted guidance for the fusion process. As a result, the fusion process can fully consider the actual conditions of image registration and the unique characteristics of different image sources, thereby significantly enhancing the quality of fusion.

As shown in Fig. 2, the forward diffusion and posterior sampling of FusDM exhibit similar characteristics to those of the RegDM. During the forward diffusion, Gaussian noise is gradually added to $\mathbf{X}^{\text{res}}$, where $\mathbf{X}^{\text{res}} = \hat{\mathbf{Z}}_{\times 4} - \mathbf{Y}_\uparrow$. When it comes to posterior sampling, at step t, the input includes $\hat{\mathbf{x}}^{\text{res}}_t$, the registered image $\hat{\mathbf{Z}}$, and $\mathbf{Y}_\uparrow$, which is specifically represented as $\mathbf{x}^{\text{in}}_{t-1} = \{\hat{\mathbf{x}}^{\text{res}}_t, \mathbf{Y}_\uparrow, \hat{\mathbf{Z}}\}$. Eventually, the residual image $\hat{\mathbf{X}}^{\text{res}}$ can be obtained. Then, $\hat{\mathbf{X}} = \mathbf{Y}_\uparrow + \hat{\mathbf{X}}^{\text{res}}$. Diffusion loss is used to optimize FusDM

$$\mathcal{L}^F_{\text{Diff}}(\omega_\psi) = \mathbb{E}\left\|\Psi_\psi\left(\mathbf{x}^{\text{in}}, t; \omega_\psi\right) - \varepsilon_f\right\|^2_2 \quad \text{s.t. } \varepsilon \sim \mathcal{N}(0, I). \tag{15}$$

To further optimize the overall performance of the model, the mean-square-error (mse) loss is employed for optimization

$$\mathcal{L}^F_{\text{mse}}(\omega_\psi) = \left\|(\mathbf{X} - \mathbf{Y}\uparrow) - \Psi_\psi\left(\mathbf{x}^{\text{in}}, t; \omega_\psi\right)\right\|^2_F. \tag{16}$$

The total loss function of the FusDM is represented as

$$\mathcal{L}^F(\omega_\psi) = \lambda\mathcal{L}^F_{\text{Diff}}(\omega_\psi) + \mathcal{L}^F_{\text{mse}}(\omega_\psi) \tag{17}$$

where λ is set to 0.00001.

D. Joint Optimization

Our proposed method introduces joint optimization loss as a conditional guidance term. The proposed method enhances the mitigation transfer between image registration and fusion, ultimately improving the overall performance of both tasks. In addition, joint learning can reduce the risk of overfitting, as the model needs to perform well on multiple tasks to achieve true optimization, as shown in Fig. 1.

Integer-order derivatives (first-order, second-order, etc.) have been widely used in image processing, such as the Sobel and Laplacian operators. The fractional-order derivative serves as a generalization of the integer-order derivative, enabling the definition of derivative orders within the field of arbitrary real numbers. The traditional integer-order derivative is relatively fixed when describing image features, and there are limitations in detail. The fractional-order derivative, however, introduces noninteger-order parameters [59]. For example, by defining the fractional-order derivative in the frequency domain [60], different fractional orders can highlight the details and structural information of an image to various extents. Compared with the integer-order derivative, it has greater flexibility in capturing the complex changes of an image. In addition, the fractional-order derivative can capture details easily overlooked by the integer-order derivative in image reconstruction and recognition. For instance, the fractional-order weighted spherical Bessel–Fourier moments (FrSBFMs) [61] can make the extraction of image features more accurate.

Assume that its domain Ω is a bounded open region. FusDM extracts the texture component of the image and employs the fractional-order derivative [62] to characterize this texture. Consider the row i and column j.

In the i direction, assume $n = \lceil\alpha\rceil$. Its α-order ($\alpha > 0$) Riemann–Liouville fractional-order derivative is

$$D^\alpha_i \hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(i, j) = \frac{1}{\Gamma(n-\alpha)}\frac{\partial i}{\partial i^n}\int_a^i \frac{\hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(\tau, j)}{(i-\tau)^{\alpha-n+1}}d\tau \tag{18}$$

where $\hat{\mathbf{X}}^{\text{res}}_{\text{CA}} = \Phi(\mathbf{z}^{\text{in}}, t; \omega_\varphi)_{\text{CA}}$. In the j-direction

$$D^\alpha_j \hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(i, j) = \frac{1}{\Gamma(n-\alpha)}\frac{\partial i}{\partial j^n}\int_a^j \frac{\hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(i, \tau)}{(j-\tau)^{\alpha-n+1}}d\tau. \tag{19}$$

Here, a represents the lower boundary of the integral. The subscript b on the right represents the upper boundary of the integral. $[a, b] \subseteq \Omega$. $\Gamma(\cdot)$ represents the gamma function.

Finally, the loss function can be expressed as a tensor product of the model, as follows:

$$\mathcal{L}_{\text{fra}}(\omega_\varphi, \omega_\psi) = \int_\Omega \left|D^\alpha_i \Phi\hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(i, j) \otimes D^\alpha_j \hat{\mathbf{X}}^{\text{res}}_{\text{CA}}(i, j)\right| d\tau - \int_\Omega \left|D^\alpha_i \hat{\mathbf{Z}}^{\text{res}}(i, j) \otimes D^\alpha_j \hat{\mathbf{Z}}^{\text{res}}(i, j)\right| d\tau \tag{20}$$

where $\hat{\mathbf{Z}}^{\text{res}} = \mathbf{X}_{\text{ca}} - \Psi(\mathbf{x}^{\text{in}}, t; \omega_\psi)$. In the proposed method, α is set to 0.5.

E. Algorithms

Algorithm 1 summarizes the training and testing processes of the overall framework for the proposed RFDifNet.

IV. Experiments

In this section, we analyze the experimental results of the proposed RFDifNet for image registration and fusion.

A. Experimental Configurations

1) Datasets: We use three multisource remote sensing datasets, Chikusei, GeoEye-1, and Pavia, to construct validation datasets, covering three types of fusion tasks: MS–HS,

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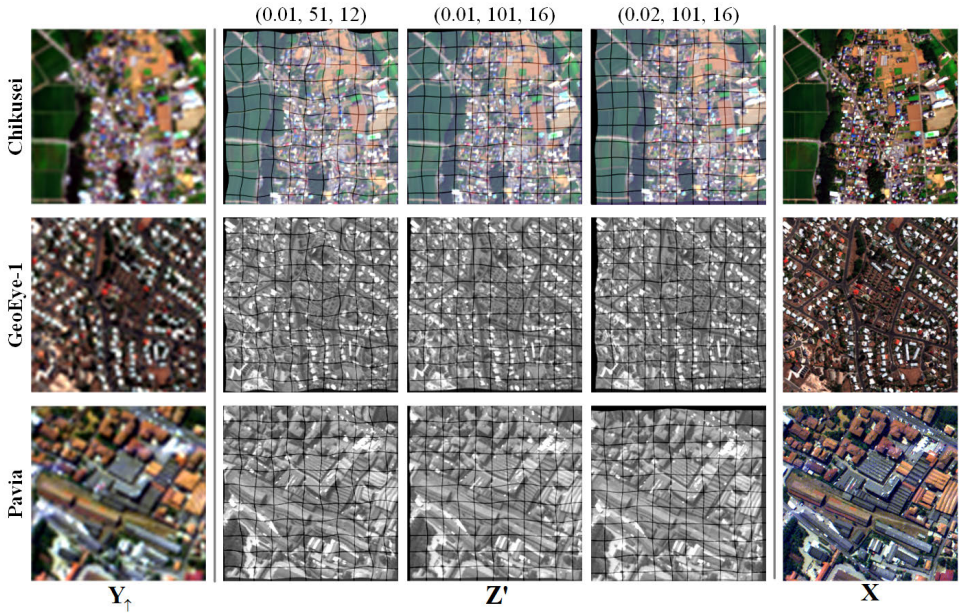


Fig. 3. Simulated datasets. According to Wald's protocol, paired simulated datasets $[\mathbf{Y}_{\uparrow}, \mathbf{Z}', \mathbf{X}]$ are obtained. Among them, $\mathbf{Z}'$ undergoes random deformation. The deformation parameter of the translation probability and the size and standard deviation of the Gaussian kernel are represented as (p, k, σ). To better display deformations, use a checkerboard pattern to calibrate the images.

Algorithm 1 Alternating Optimization for the RFDifNet (Image Registration and Fusion)

TRAINING PHASE

Input: Unregistered $\mathbf{Z}' \in \mathbb{R}^{W\times H\times l}$, upsampling $\mathbf{Y}_{\uparrow} \in \mathbb{R}^{W\times H\times L}$, and reference $\mathbf{X} \in \mathbb{R}^{W\times H\times L}$;

Initialize parameter $\Phi\left(\omega_{\varphi}\right)$ of RegDM and parameter $\Psi\left(\omega_{\psi}\right)$ of FusDM;

for iteration = 1 to 300000 **do**

Forward Propagation:

1. Generate registered images: $\hat{\mathbf{Z}} = \Phi(\phi_0^{-1}, \mathbf{Z}', \mathbf{Y}_{\uparrow}; \omega_{\varphi})$
2. Generate fused images: $\hat{\mathbf{X}} = \Psi(\hat{\mathbf{Z}}, \mathbf{Y}_{\uparrow}; \omega_{\psi})$
3. Compute losses $\mathcal{L}^R$, $\mathcal{L}^F$, and $\mathcal{L}_{fra}$

Backward Propagation (Alternating Update):

1. Fix ω_{ψ}, update ω_{φ} to minimize $\mathcal{L}^R$,optimize RegDM
2. Fix ω_{φ}, update ω_{ψ} to minimize $\mathcal{L}^F$,optimize FusDM
3. Extract residual features via fractional derivatives $\mathcal{L}_{fra}$ for fine correction

if iteration $\equiv$ 0 mod 5000 **then**

Evaluate model on the validation set

Save best-performing parameters

end if

end for

TESTING PHASE

Input: Unregistered $\mathbf{Z}' \in \mathbb{R}^{W\times H\times l}$ and upsampling $\mathbf{Y}_{\uparrow} \in \mathbb{R}^{W\times H\times L}$;

1. Generate registered output: $\hat{\mathbf{Z}} = \Phi(\phi_0^{-1}, \mathbf{Z}', \mathbf{Y}_{\uparrow}; \omega_{\varphi}^*)$
2. Generate fused output: $\hat{\mathbf{X}} = \Psi(\hat{\mathbf{Z}}, \mathbf{Y}_{\uparrow}; \omega_{\psi}^*)$

Output: Registered images $\hat{\mathbf{Z}} \in \mathbb{R}^{W\times H\times l}$ and fused images$\hat{\mathbf{X}} \in \mathbb{R}^{W\times H\times L}$;

MS–PAN, and HS–PAN. Table I shows that the GeoEye-1 dataset contains a real dataset that maintains its original

TABLE I
DETAILS OF THE DATASETS USED IN THE FOLLOWING EXPERIMENTS

sensor	Spatial resolution(m)		Image size	
	Y	**Z**	**Y**	**Z**
Chikusei	10	2.5	64×64×128	256×256×4
GeoEye-1(sim.)	2	0.5	64×64×4	256×256×1
GeoEye-1(real)	0.5	0.125	64×64×4	256×256×1
Pavia	5.2	1.3	64×64×102	256×256×4

imaging characteristics and simulated datasets based on Wald's protocol. The datasets generate paired samples $[\mathbf{Y}_{\uparrow}, \mathbf{Z}, \mathbf{X},]$. Here, $\mathbf{Y}_{\uparrow}$ denotes the low-spatial-resolution image, $\mathbf{Z}$ represents the high-spatial-resolution image, and $\mathbf{X}$ is the reference image with both high spatial and spectral resolutions. The Chikusei and Pavia datasets also use simulated data based on Wald's protocol. Fig. 3 specifically shows the simulated deformation images $\mathbf{Z}'$.

We choose the HR image $\mathbf{Z}$ for transformation [50]. It contains abundant, detailed information, which can provide a more accurate basis for simulating various complex distortions. We apply varying degrees of random affine and elastic transformations to generate multiple deformation fields, which are used to $\mathbf{Z}$ to simulate various distortions to obtain $\mathbf{Z}'$. This process enables us to evaluate the robustness and performance of the proposed RFDifNet under different conditions.

The image registration in the proposed RFDiNet aims to address two types of geometric differences: rigid (translational) and nonrigid (elastic deformation). Among them, rigid transformations simulate sensor offset or platform stability issues (such as translation of 1%–2%). In contrast, nonrigid transformations simulate terrain undulations or sensor distortions (such as elastic deformation with a Gaussian kernel $\sigma \in \{12, 16\}$, $k \in \{51, 101\}$), where σ denote standard deviation and k denotes size. The experiment shows that the proposed

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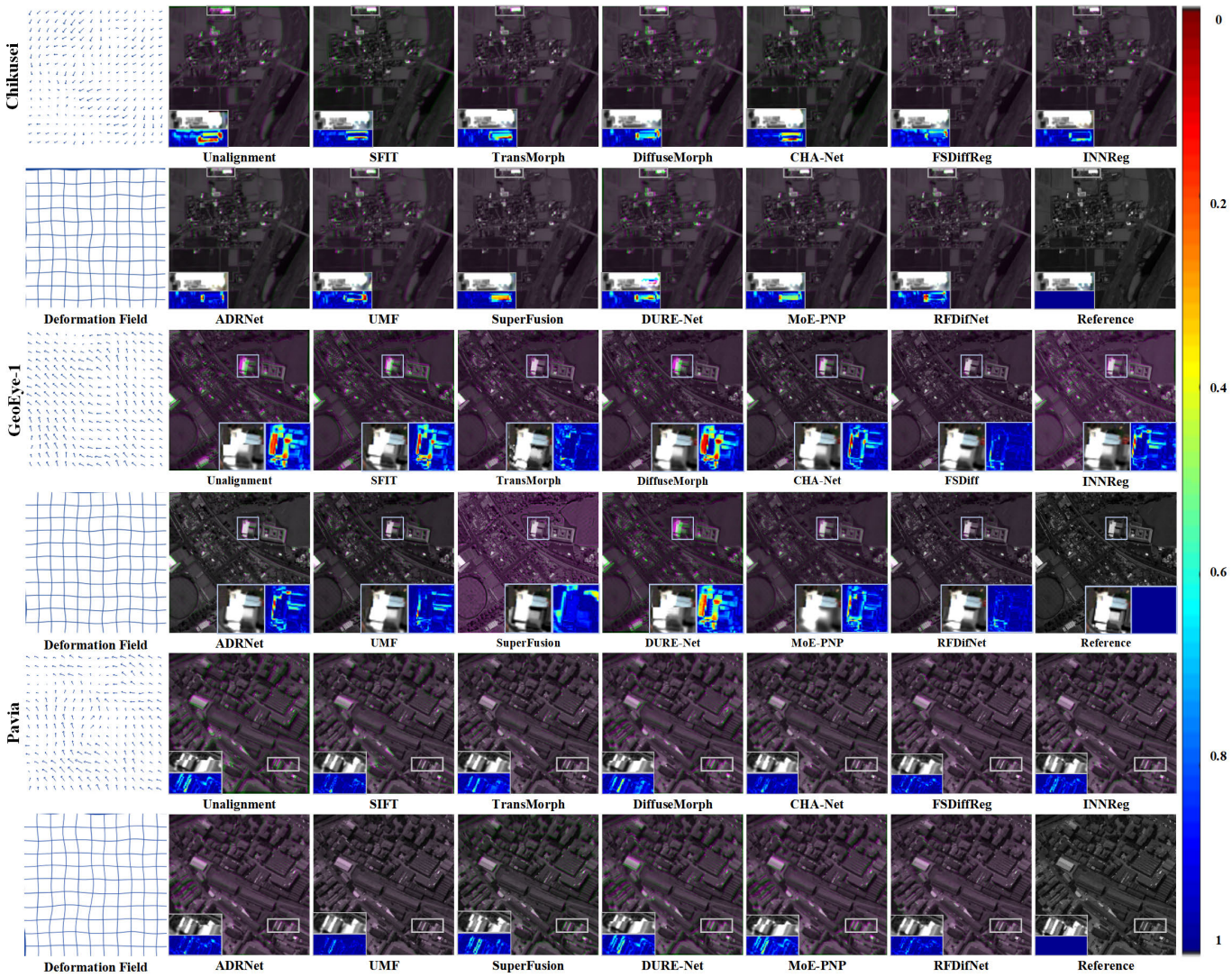


Fig. 4. Quantitative comparison on the registered images of the simulated datasets. The optical flow maps in the form of an arrow and grid in the first column are used to indicate the direction and size of the deformation field. Differences between the registration results of the comparison method and RFDifNet are compared to the reference image. Additionally, regions with significant changes are highlighted using checkerboard comparison and residual maps.

TABLE II

QUALITATIVE COMPARISON ON THE REGISTERED IMAGES OF THE SIMULATED DATASETS

		MS and HS (Chikusei)		MS and PAN (GeoEye-1)		HS and PAN (Pavia)	
		MSE	NCC	MSE	NCC	MSE	NCC
	Index	0	1	0	1	0	1
	Unregistered input	0.001798	0.8185	0.000592	0.5687	0.002792	0.6699
Registration	SIFT'04	0.001551	0.8290	0.000287	0.7292	0.001792	0.7872
	TransMorph'22	0.001083	0.8826	0.000371	0.7390	**0.000492**	0.8511
	DiffuseMorph'22	0.001462	0.8510	0.000220	0.8316	0.004056	0.7565
	CHA-Net'23	0.000941	0.9039	0.000245	0.8132	0.001766	0.7856
	FSDiffReg'23	0.001820	0.9320	0.000201	0.8496	0.000945	0.8863
	INNReg'24	0.000878	0.9092	0.000359	0.7228	0.001406	0.8311
	ADRNet'24	0.000644	0.9360	0.000211	0.8422	0.001394	0.8318
Registration & Fusion	UMF'22	0.001466	0.8498	0.000274	0.7846	0.001291	0.8431
	SuperFusion'22	0.001436	0.8439	0.000413	0.8076	0.001845	0.7847
	DURE-Net'23	0.001741	0.8188	0.000590	0.5698	0.002792	0.6699
	MoE-PNP'24	0.001159	0.8767	0.000235	0.8260	0.001429	0.8272
	RFDifNet	**0.000511**	**0.9526**	**0.000180**	**0.8628**	0.000653	**0.9260**

method performs best under moderate deformation, as shown in Fig. 8, while excessive deformation may lead to local information loss. It is necessary to balance accuracy and robustness in application scenarios.

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TABLE III
QUALITATIVE COMPARISON ON THE FUSED IMAGES OF THE SIMULATED DATASETS

			CC	RMSE	ERG	SAM	Q4	UIQI
			1	0	0	0	1	1
MS and HS (Chikusei)	Fusion	HySure'15	0.9148	198.6656	0.4202	3.3016	-	0.9062
		MoG-DCN'21	0.9313	158.5303	0.4831	2.4700	-	0.9230
		MSST-Net'23	0.9544	143.4591	0.3471	2.5606	-	0.9439
		PSRT'23	0.9638	**119.1410**	0.3231	2.0458	-	0.9549
		DDIF'24	0.9287	177.9957	0.4950	2.9286	-	0.9193
	Registration & Fusion	UMF'22	0.8206	284.9865	0.6700	3.9224	-	0.8121
		SuperFusion'22	0.8795	220.0374	0.6046	3.4724	-	0.8635
		DURE-Net'23	0.9455	145.4176	0.4091	2.3671	-	0.9356
		PMI-RFCoNet'24	0.9238	179.9591	0.4829	2.9756	-	0.9144
		MoE-PNP'24	0.9070	185.3716	0.5451	3.2263	-	0.8964
		RFDifNet	**0.9641**	120.4258	**0.3121**	**1.7715**	-	**0.9555**
MS and PAN (GeoEye-1)	Fusion	BDSD'08	0.9706	39.6671	2.5717	6.1876	0.6955	0.8730
		PSGAN'20	0.8735	23.0863	1.4904	3.8798	0.7968	0.9550
		UCGAN'22	0.9561	37.0997	2.3983	5.7866	0.6826	0.8657
		HyperTransformer'22	0.8855	48.0020	3.8143	4.5217	0.7400	0.8736
		DDIF'24	0.8847	46.6805	1.2257	3.1532	**0.8122**	0.9701
	Registration & Fusion	UMF'22	0.8726	38.4483	2.5063	4.8788	0.6714	0.8628
		SuperFusion'22	0.8902	35.1752	2.4044	6.6606	0.7066	0.8828
		DURE-Net'23	0.9682	19.7921	1.2743	3.4855	0.8089	0.9675
		PMI-RFCoNet'24	0.9600	22.2323	1.4324	3.6851	0.7987	0.9596
		MoE-PNP'24	0.9498	24.9575	1.6121	4.1200	0.7965	0.9492
		RFDifNet	**0.9711**	**18.9184**	**1.2191**	**3.1288**	0.8115	**0.9708**
HS and PAN (Pavia)	Fusion	HyperPNN'19	0.9566	189.5708	0.3986	4.8823	-	0.9453
		DHP-DARN'20	0.9295	151.4023	0.5173	5.2101	-	0.9174
		DIP-HyperKite'22	0.9604	181.8115	0.3833	4.7284	-	0.9494
		FPFNet'23	0.9566	189.5708	0.3986	4.8823	-	0.9453
		DDIF'24	0.9594	**112.9270**	0.3894	3.9318	-	0.9496
	Registration & Fusion	UMF'22	0.8636	408.4063	0.9157	9.6814	-	0.8214
		SuperFusion'22	0.7819	424.7214	0.8981	7.7316	-	0.7284
		DURE-Net'23	0.9289	242.2067	0.5227	5.3461	-	0.9178
		PMI-RFCoNet'24	0.9307	240.0889	0.5142	5.2673	-	0.9209
		MoE-PNP'24	0.8765	317.9299	0.6730	6.0543	-	0.8564
		RFDifNet	**0.9624**	174.3967	**0.3765**	**3.7845**	-	**0.9526**

TABLE IV
QUANTITATIVE COMPARISON ON THE REGISTERED IMAGES OF THE REAL GEOEYE-1 DATASETS

		MSE	NCC
	Index	0	1
	Unregistered input	0.00026124	0.6818
Registration	SIFT'04	0.00020570	0.7358
	TransMorph'22	0.00021527	0.7414
	DiffuseMorph'22	0.00018985	0.7566
	CHA-Net'23	0.00013523	0.8152
	FSDiffReg'23	0.00051137	0.8568
	ADRNet'24	0.00024623	0.6916
	INNReg'24	0.00010054	0.8422
Registration & Fusion	UMF'22	0.00015553	0.7818
	SuperFusion'22	0.00260000	0.5726
	DURE-Net'23	0.00025912	0.6840
	MoE-PNP'24	0.00008313	0.8534
	RFDifNet	**0.00008250**	**0.8600**

According to (2), rigid transformations involve translation, where the resulting coordinates are defined by $i' = i + u$ and $j' = j + v$, with u and v representing the translation amounts in the horizontal and vertical directions, respectively. The probabilities of displacement were set at 1% and 2%. Nonrigid transformations involve random elastic deformations, characterized by a random noise field $\vec{n}(i, j) = [n_1(i, j), n_2(i, j)]$, which is smoothed using a Gaussian kernel $G(i, j)$ to obtain the displacement field $\vec{d}(i, j) = [n_1(i, j) * G(i, j), n_2(i, j) * G(i, j)]$, leading to the postelastically deformed coordinates $i' = i + n_1(i, j) * G(i, j)$ and $j' = j + n_2(i, j) * G(i, j)$. Here, $*$ denotes the convolution operation. Smaller kernels and standard deviations result in more significant deformations.

2) Comparison Method: The following joint optimization methods for registration and fusion are compared: UMF [63], SuperFusion [64], DURE-Net [65], PMI-RFCoNet [33], and MoE-PNP [32].

Due to the limited number of joint optimization methods available for registration and fusion, we first register the image $\mathbf{Z}'$ using state-of-the-art registration methods, then use the best registration result from the comparison as the input image for the fusion methods. The registration methods include SIFT [56], TransMorph [51], DiffuseMorph [52], CHA-Net [50], FSDiffReg [53], INNReg [66], and ADRNet [67]. Based on

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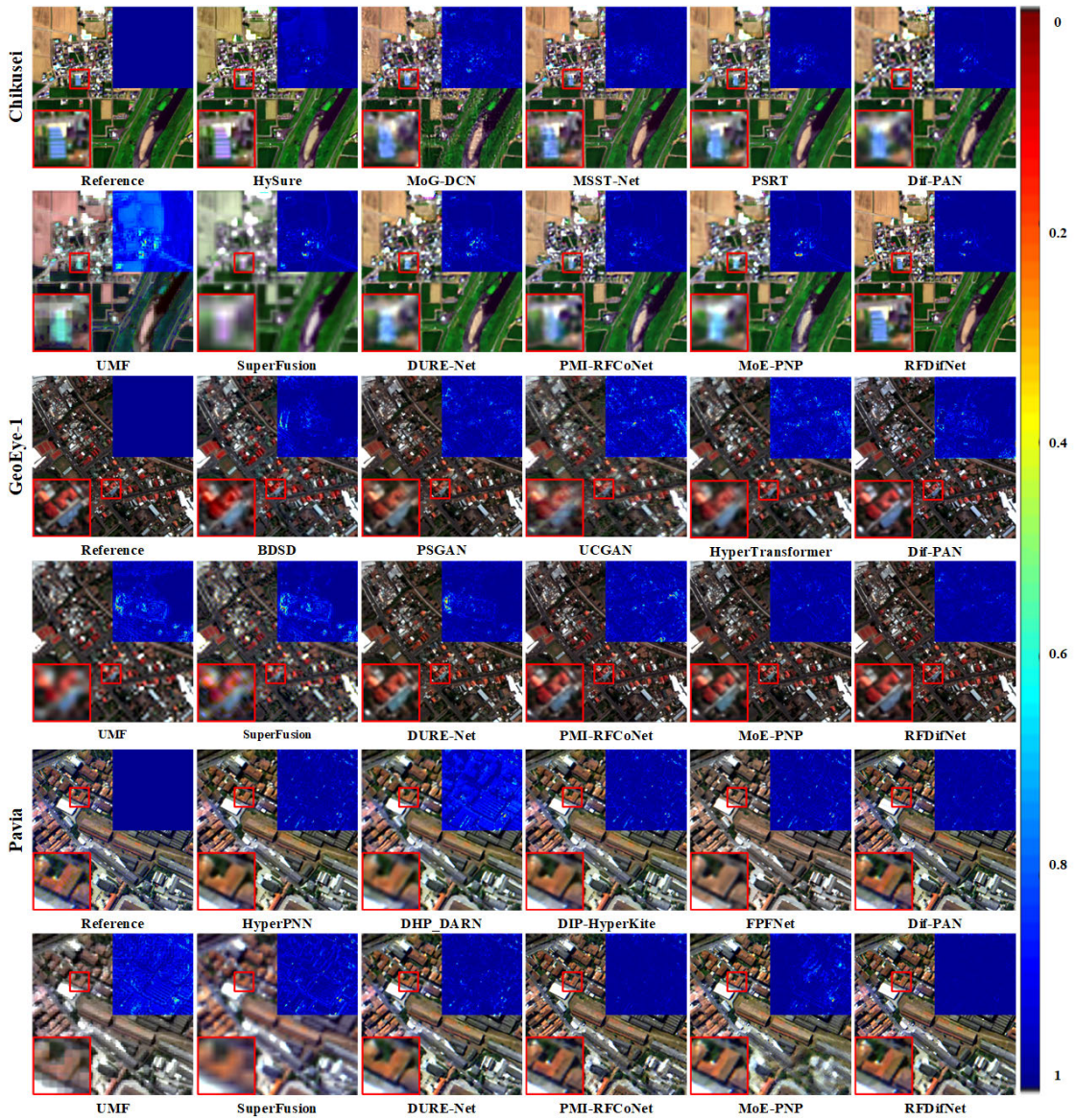


Fig. 5. Quantitative comparison on the fused images of the simulated datasets. Pseudocolor images of the fusion results from the comparison method and RFDifNet. The band combinations used are as follows: Chikusei (R:61, G:35, B:10), GeoEye-1 (R:3, G:2, B:1), and Pavia (R:57, G:34, B:3). The residual map between the fused image and the reference image is shown in the upper right corner. Additionally, an enlarged view of a complex area is provided to illustrate the generated details.

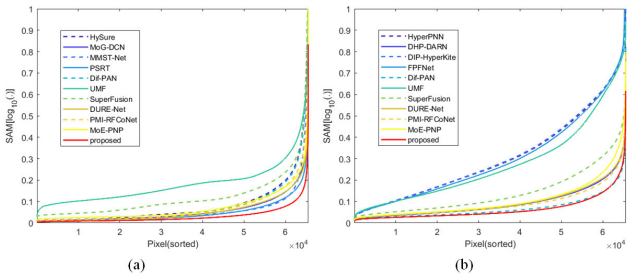


Fig. 6. SAM (plotted on a log10(•) scale) as a function of sorted pixels. (a) Chikusei. (b) Pavia.

subsequent comparison experiments, the registration result from FSDiffReg is selected as the input for the fusion methods.

The comparison methods employed for the fusion of MS and HS images include HySure [68], MoG-DCN [25], MSST-Net [31], PSRT [30], and DDIF [23]. For the fusion of MS and PAN images, the methods utilized are BDSD [69], PSGAN [18], UCGAN [20], HyperTransformer [21], and DDIF. In the context of the HS and PAN images, the comparison methods consist of HyperPNN [42], DHP-DARN [41], DIP-HyperKite [40], FPFNet [43], and DDIF. Dif-PAN [23] is a state-of-the-art fusion method based on diffusion models. We have implemented adaptive improvements to the DDIF [23] to achieve optimal performance for each task.

3) Implementation Details: The RFDifNet was trained and tested on an NVIDIA GeForce RTX 3090 GPU and an Intel[1] Core[2] i7-9700KF CPU @ 3.60 GHz. The RFDifNet employs the Adam optimizer, with learning rates for the registration and fusion modules set at 0.0002 and 0.0001, respectively. The weight of $\mathcal{L}_{\text{fra}}$ was set to 0.001.

[1]Registered trademark.
[2]Trademarked.

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TABLE V
QUANTITATIVE COMPARISON ON THE FUSED IMAGES OF THE REAL GEOEYE-1 DATASETS

		D_r	D_s	QNR
	Index	0	0	1
Fusion	BDSD'08	0.0625	0.1775	0.7716
	PSGAN'20	0.0856	0.0978	0.8241
	UCGAN'22	0.0684	0.1979	0.7478
	HyperTransformer'22	0.0559	0.1403	0.8108
	DDIF'24	0.0658	0.1124	0.8286
Registration & Fusion	UMF'22	0.0668	0.1675	0.7774
	SuperFusion'22	0.0869	0.2092	0.7233
	DURE-Net'23	**0.0522**	0.0951	0.8578
	PMI-RFCoNet'24	0.0562	0.0899	0.8590
	MoE-PNP'24	0.0530	0.0931	0.8590
	RFDifNet	0.0529	**0.0886**	**0.8631**

TABLE VI
COMPARISON OF PARAMETERS, INFERENCE TIME, NCC, AND SAM (PAVIA). TEST IMAGE SIZE: $1 \times 1 \times 256 \times 256$

Task	Method	Params (MB)	Time (ms)	NCC	SAM
Registration	SIFT'04	-	111.01	0.7872	-
	TransMorph'22	8.0	40.00	0.8511	-
	DiffuseMorph'22	60.82	328.53	0.7565	-
	CHA-Net'23	27.28	75.56	0.7856	-
	FSDiffReg'23	1.79	113.62	0.8863	-
	INNReg'24	44.74	180.11	0.8311	-
	ADRNet'24	288.33	75.28	0.8318	-
Fusion	HyperPNN'19	0.51	56.91	-	4.8823
	DHP-DARN'20	1.59	64.89	-	5.2101
	DIP-HyperKite'22	2.00	219.67	-	4.7284
	FPFNet'23	179.81	1036.61	-	4.8823
	DDIF'24	52.79	3061.70	-	3.9318
Reg.&Fusion	UMF'22 (Reg)	0.65	10.02	0.8431	9.6814
	UMF'22 (Fus)	2.85	17.43		
	SuperFusion'22	7.59	67.36	0.7847	7.7316
	DURE-Net'23	2.77	83.11	0.6699	5.3461
	PMI-RFCoNet'24	64.84	84.21	-	5.2673
	MoE-PNP'24	173.20	284.44	0.8272	6.0543
	RFDifNet (Reg)	5.85	764.11	0.9260	3.7845
	RFDifNet (Fus)	50.15	3836.69		

We employ mse and NCC to assess the registration accuracy between the registered and reference images for the quantitative evaluation of image registration. For image fusion, the following metrics: correlation coefficient (CC), root-mean-squared error (RMSE), *Erreur Relative Globale Adimensionnelle de Synthèse* (ERGAS) [70], spectral angle mapper (SAM) [71], and universal image quality index (UIQI) [72]. Additionally, Q4 [73] is used as a quantitative evaluation metric for pan-sharpening. The best values for each of the quantitative evaluation metrics are presented in Tables II and III, specifically.

B. Results on Image Registration

Fig. 4 shows the qualitative evaluation of registration performance between the RFDifNet and existing comparison methods on three simulated datasets (Chikusei, GeoEye-1, and Pavia). The parameters (p, k, σ) are set to $(0.01, 101, 16)$. We adopted the pseudocolor synthesis method to highlight the differences between the registered and reference images. This method maps the intensity differences of corresponding pixel positions between two images to different colors, thereby visually emphasizing registration errors. Specifically, red represents higher intensity in the registered image, while green represents higher intensity in the reference image. In addition, the figure also includes an enlarged view of the selected regions to provide a more detailed comparison of the registration and residual maps between the registered and reference images. Fig. 4 shows that the RFDifNet achieves the best registration results, displaying the highest registration accuracy and minimum residuals. This indicates that the proposed method has significant advantages in handling complex geographic spatial data.

Table III summarizes the quantitative evaluation metrics of the RFDifNet and the comparison methods of the three datasets. The statistical results show that the proposed method consistently outperforms the comparison methods in all evaluation criteria, highlighting its superior performance in image registration. It is worth noting that FSDiffReg, also based on the diffusion model, achieved excellent registration results, further verifying the effectiveness of the diffusion model in remote sensing image processing.

C. Results on Image Fusion

Fig. 5 presents a qualitative assessment of the image fusion results for the RFDifNet compared to other methods on three simulated datasets: Chikusei, GeoEye-1, and Pavia. To ensure registration accuracy, we employed the high-performing registration method FSDiffReg to preprocess the original images. The resulting registered images, denoted as $\hat{\mathbf{Z}}$, were used as input for all fusion methods, ensuring consistent experimental conditions. The upper right corner of Fig. 5 displays the residual image between the fused and reference images. The residual image provides a clear visualization of the reconstruction error for each method. In addition, to compare the performance of different methods in more detail, we selected complex regions in the image and enlarged the regions. It is evident from these visualization results that the RFDifNet not only achieves the highest quality visual fusion but also produces the lowest residual values. Overall, the proposed method has an outstanding ability to preserve details and reduce artifacts.

The quantitative evaluation indicators of the RFDifNet and the comparison methods of Chikusei, GeoEye-1, and Pavia datasets are summarized in Table III. The statistical results indicate that the proposed method outperforms the comparative method on almost all evaluation criteria. It is worth noting that the DDIF [23] based on the diffusion model has achieved excellent performance, further verifying the potential and effectiveness of the diffusion model in remote sensing image fusion.

We evaluated the performance of the RFDifNet using pixel-by-pixel SAM curves on the HS dataset Chikusei and Pavia. As shown in Fig. 6, our method exhibits the highest average

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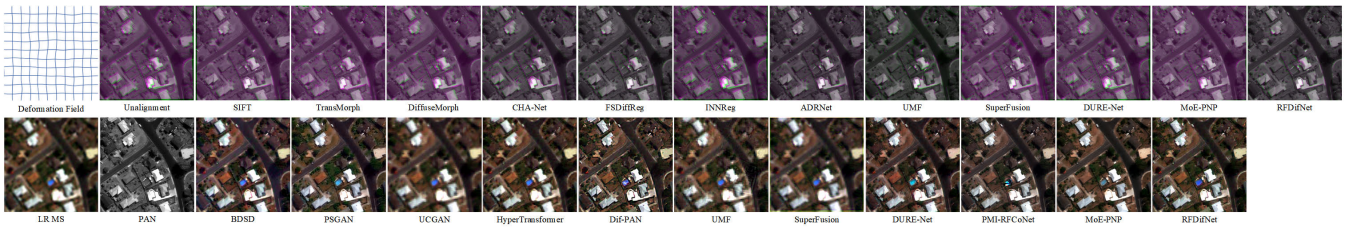


Fig. 7. Qualitative comparison on the registered and fused images of the real GeoEye-1 datasets.

spectral fidelity on all pixels, highlighting the effectiveness of the diffusion generation process.

D. Real Datasets

To evaluate the RFDifNet more comprehensively, we conducted detailed qualitative and quantitative analyses using the real GeoEye-1 dataset, with the results presented in Fig. 7 and Tables IV and V. The parameters (p, k, σ) are set to $(0.01, 101, 16)$. As shown in Fig. 7, the RFDifNet demonstrates superior performance in accurately aligning image features, preserving spatial structures, and providing realistic spectral characteristics, thereby improving the overall quality of images. Tables IV and V further demonstrate that the RFDifNet achieves excellent spatial fidelity. The proposed method has reliability and effectiveness in practical applications.

E. Analysis of Different Distortions

Fig. 8 presents the qualitative and quantitative analysis results of the simulated datasets after applying different transformations. According to the data in Fig. 8, when the parameters (p, k, σ) are set to $(0.01, 101, 16)$, the RFDifNet exhibits reduced geometric and appearance differences compared to the comparison methods within the simulated dataset. Given the requirements for subsequent fusion, achieving the most precise registration possible is crucial. In addition, while larger deformations provide greater flexibility, they also increase computational costs. In practical applications, excessive deformation can lead to the loss of partial image data. It is necessary to select appropriate deformations based on specific application scenarios for verification to determine the optimal parameters.

As shown in Fig. 8, when the translation parameter is 1% and the elastic deformation parameters are $(\sigma = 16, k = 101)$, this method significantly reduces fusion artifacts while maintaining geometric consistency. Although larger deformations can increase the challenge, they can lead to loss of edge information (such as blurred building contours), affecting the quality of subsequent fusion. Therefore, the deformation parameters selected in this article balance practical needs and algorithm stability and align in most remote sensing application scenarios.

F. Ablation Study

1) Comparison Between Fractional Order and Integer Order: Fig. 9 presents the differences in quantitative analysis between fractional-order and integer-order derivative losses. Among them, "none" represents the situation where the joint optimization loss strategy is not applied. For integer-order losses, the first-order integer Sobel and the second-order integer Laplacian operators are employed for calculations. It is obvious that fractional-order derivative loss exhibits significantly enhanced effects in optimizing RegDM performance by comparison. Compared with integer-order derivative losses, fractional-order losses offer substantial advantages and contribute to more pronounced performance improvements for the RegDM.

It is worth noting that when the parameter value is set to 0, the system reverts to a two-stage process: an independent registration step followed by an independent fusion step. This configuration means that the fusion result does not influence the registration. The results shown in Fig. 9 indicate that introducing joint optimization loss significantly enhances the overall performance of the framework. This indicates that $\mathcal{L}_{\text{fra}}$ can effectively coordinate the two tasks, enhancing both registration and fusion performance. Therefore, it provides better control over the registration and fusion processes, enhancing the overall framework performance.

2) Evaluation of Registration and Fusion Performance Under Varying Loss Weights: Fig. 10 shows the quantitative analysis for registration and fusion results under different loss weights γ of $\mathcal{L}_{\text{fra}}$. To present this data more clearly, a dual Y-axis design is employed. As shown in Fig. 10(a), the left Y-axis represents mse, which measures the accuracy of registration. The right Y-axis represents NCC, reflecting the similarity between registered and reference images. As shown in Fig. 10(b), the left Y-axis indicates RMSE, assessing the difference between the fused image and the ground truth. The right Y-axis denotes SAM, which evaluates the fidelity of spectral information. The X-axis shows the range of parameter values, including [0, 0.1, 0.01, 0.001, 0.0001], with a logarithmic scale employed to accurately represent data spanning multiple orders of magnitude. The joint optimization loss function exhibits the best promotion effect in both registration and fusion when the parameter value is set to 0.001. At this point, the model can simultaneously improve registration accuracy and fusion quality.

G. Efficiency Experiments

In this section, we compared the parameter counting and inference time of the proposed method with other registration and fusion comparison methods. The experiment used ten image pairs (size of $256 \times 256 \times 1$) from the Pavia simulated test dataset as inputs, with the results shown in Table VI. Since

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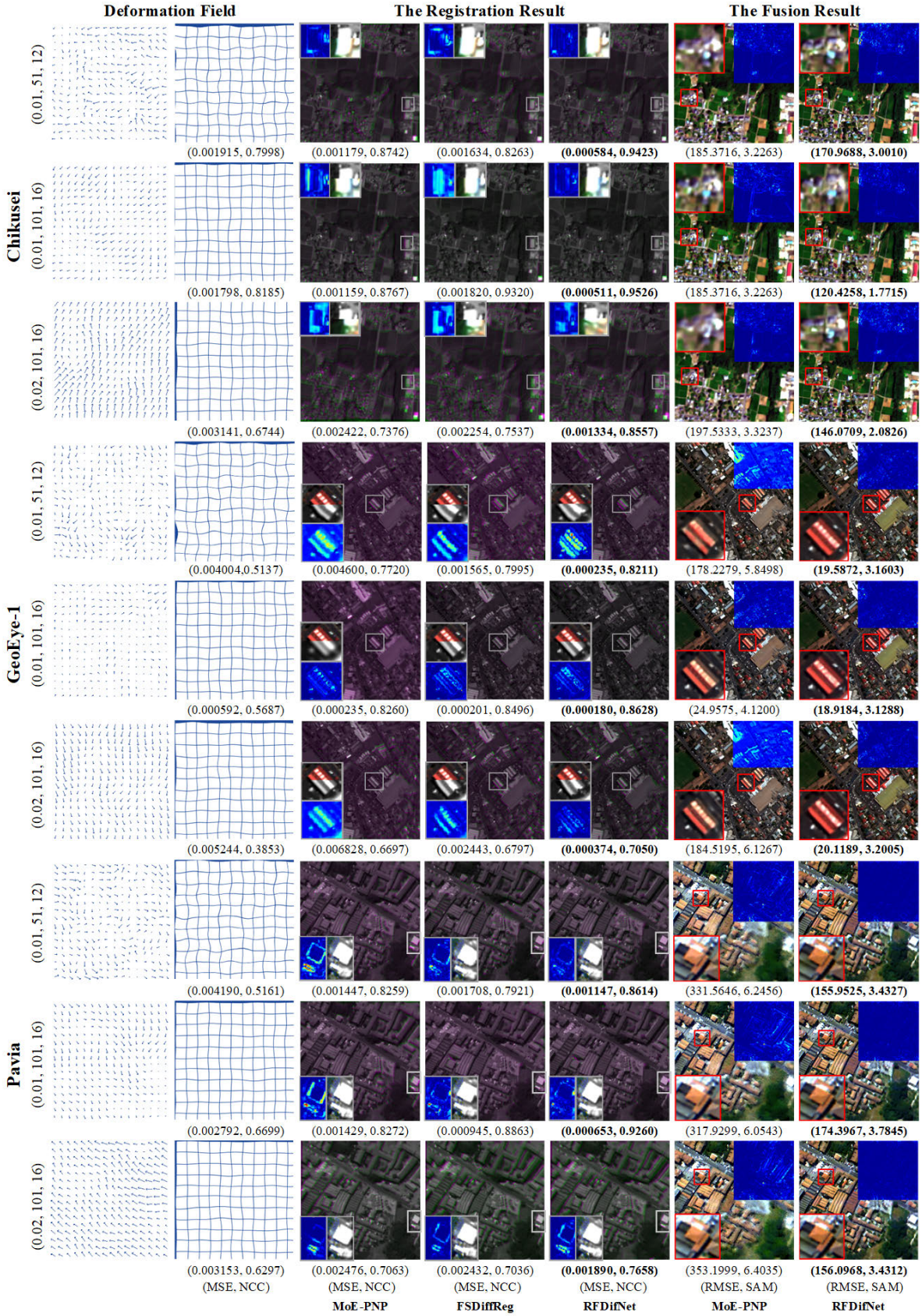


Fig. 8. Quantitative comparison on the impacts of different deformations on the registered and fused images of the simulated datasets. The best quantitative results are highlighted in red.

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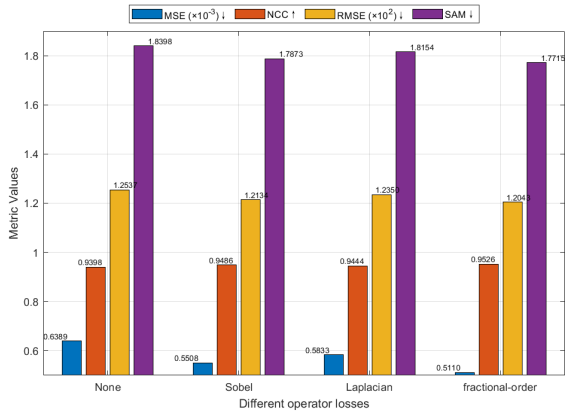
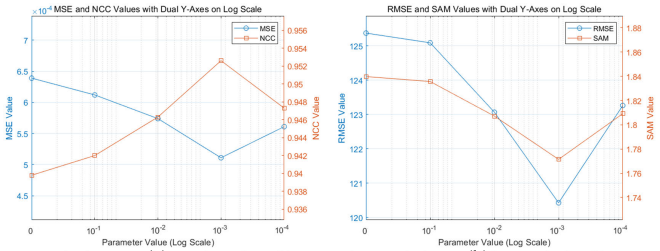


Fig. 9. Quantitative comparison on the impacts of different joint optimization operators on the registered and fused images of the simulated Chikusei datasets.



(a) (b)

Fig. 10. Quantitative comparison on the impacts of the weights assigned to the fractional-order loss on the registered and fused images of the simulated Chikusei datasets. (a) Quantitative comparison of the registered images. (b) Quantitative comparison of the fused images.

the inference process requires a time step of T for testing, the proposed method requires a long time for inference. The sampling inference stage of the proposed method uses the same DDIM method as DDIF [23] to accelerate the sampling process, and the sampling step is 25. The inference time of the proposed method is tens to hundreds of times longer than other comparison methods. Future research may improve this deficiency by accelerating inference through hybrid sampling algorithms.

V. Conclusion

The proposed method, RFDifNet, tries to solve the critical tasks of image registration and fusion under multisource remote-sensing images. The joint learning method with diffusion models is established by the interaction between two modules involved within one unified framework: registration and fusion. A closed-loop iterative mechanism is established. This interplay allows for an increase in the quality and accuracy of both the registration and fusion. Furthermore, the designed joint learning loss based on the fractional-order derivative is better than the integer-order derivative in preserving geometry. Although achieving state-of-the-art performance, the current framework has limitations, including the inherent computational overhead of iterative diffusion processes. Future research will focus on developing accelerated sampling algorithms to improve computational efficiency and combining physical models to enhance cross-modal adaptability.

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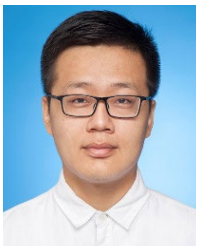
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